

Arnav Sinha

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Education

Vellore Institute of Technology

B.Tech, Computer Science and Engineering (AI/ML) | CGPA: 8.56/10.0

Vellore, India

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Software Engineering, Machine Learning, Deep Learning

Experience

Inadev Corporation

Data Science Intern

May 2026 – Aug 2026

Kolkata, India (Hybrid)

SOW AI Extraction Service – Python, FastAPI, Pydantic, Docker, AWS Lambda, S3, AWS Bedrock, Tesseract OCR

- Shipped a stateless REST microservice as a container image on AWS Lambda behind a load balancer, automating the portal's revenue-tracking workflow by turning contract PDFs from S3 into a total value and dated billing schedule; extraction failures return HTTP 200 with a FAILED status the caller can persist
- Productionized a colleague's notebook prototype in 2 days (FastAPI, Pydantic-validated schemas, one Docker image for local, ECS and Lambda) and kept every total, date and compounding step in deterministic Python, matching a 3-year 5%-escalating contract's own schedule to the rupee
- Root-caused silent revenue loss where routing pages by character count let a scanned pricing page skip OCR; fixed it by OCR-ing every page, flagged unquantifiable contracts for human review with reasons instead of guessed figures, and wrote up the RCA the same morning

Legal Contract RAG Assistant – LangGraph, MongoDB Atlas Vector Search, AWS Bedrock, LangSmith, Streamlit

- Built a citation-backed RAG knowledge assistant over 48 legal contracts on MongoDB Atlas Vector Search, with a resumable ingestion pipeline that survives API quota errors and a provider-agnostic Streamlit chat UI over Ollama, Bedrock and Gemini
- Replaced a single-shot chain with a LangGraph tool-calling agent that recovered 23/35 measured retrieval failures, and added a regression suite graded by an independent local model that caught a 26% regression, which I traced and halved

Projects

FinVoice – AI-Powered Financial Call Analytics

Python, FastAPI, SQLite, Next.js, TypeScript, GitHub Actions

- Built the FastAPI backend (21 routes) for an on-device call-analytics pipeline: background job processing behind a GPU semaphore, a SQLite call index, API-key auth and CSV/Parquet exports, plus a Next.js dashboard with live progress and an approve/escalate/reject review queue
- Added per-stage degradation tracking and timing instrumentation that surfaced silent failures and made the slowest analyzer $4.7\times$ faster (73.9s to 15.6s) with no lost signals; 139 tests and GitHub Actions CI

Smart Operator Assistant – Caterpillar Hackathon

Python, FastAPI, SQLite, Next.js, LightGBM

- Built an offline-first FastAPI + SQLite backend (54 endpoints, WAL mode) for an in-cab machine-operator assistant: an idempotent job state machine with a command ledger, alert arbitration that logs every suppression, and a Next.js touchscreen UI
- Embedded LightGBM job-time forecasts with calibrated 90% ranges and One-Class SVM anomaly alerts behind the API; 414 tests and 1–5 ms median local API latency

Ripple-Aware Satellite Collision Avoidance – Capstone Research

Python, JAX, Multi-Agent RL, SGP4

- Built a GPU-vectorized orbital simulator in JAX (two-body + J2 gravity, 837 real objects) running 14.4M object-steps/s on a laptop GPU with 44 ms conjunction-screening passes, validated by 16 physics and screening tests
- Processed 620K real Space-Track orbit records with a drag-aware manoeuvre detector, showing every manoeuvre makes the public catalogue $2.2\times$ less accurate; trained a MAPPO + GNN policy on the simulator

Publications & Achievements

- R. Bansal, S. Madan, **A. Sinha**, Karthik G M, "Ripple-Aware Cooperative Collision Avoidance for Satellite Mega-Constellations," accepted at *ICoN INDIA 2026* (IEEE technically co-sponsored), Pondicherry University
- First Place, IEEE CS VIT Hackbattle 2025** (600+ participants): built TheBlinkingWorlds, a 2D platformer in Godot where real-time OpenCV blink detection switches the game world

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Backend & Databases: FastAPI, Flask, Node.js, REST APIs, Pydantic, PostgreSQL, MySQL, SQLite, MongoDB (Atlas Vector Search)

Cloud, DevOps & Tools: AWS (Lambda, EC2, S3, Bedrock), Docker, GitHub Actions, Git/GitHub, Linux, pytest, React.js, Next.js, Streamlit

AI/ML: PyTorch, JAX, scikit-learn, LangGraph, LangChain, RAG, Vector Search, HuggingFace Transformers, OpenCV